2026–2027 AIDaS Seed Grant Program

Self-Repairing AI Agents on the Wayne State Grid: From Labeled CT to C-Arm Level Localization, with Medical Trainees Driving the Agents

Strategic areas	1, Foundational AI (primary): an agentic system held to verifiable gates. 4, AI for Health: medical imaging for intraoperative decision support. 3, AI for Education and Workforce Training: medical trainees doing computational research. Aim 1 couples AI to the university's high-performance computing, which also serves area 2, AI for Science.
Principal Investigator	Zhizhong Han, PhD, Assistant Professor, Department of Computer Science, College of Engineering, Wayne State University; founding director of the Machine Perception Lab; AIDaS affiliated member. Contact: [DATA NEEDED: office, phone and e-mail].
Co-Principal Investigator	Loren Schwiebert, PhD, Professor, Department of Computer Science, College of Engineering, Wayne State University; AIDaS affiliated member. Contact: [DATA NEEDED: office, phone and e-mail].
Clinical collaborator	Gregory Schwing, MD, PhD, General Surgery, Wayne State University School of Medicine / Detroit Medical Center. Receives no award funds.
New Investigator status	The PI is a New Investigator: an Assistant Professor at Wayne State since January 2021 [DATA NEEDED: confirm that the appointment is tenure-track], and an NIH Early Stage Investigator until November 2027, ten years after the PI's PhD of November 2017 [DATA NEEDED: confirm the end date in eRA Commons]. The Co-PI is a Professor and not a New Investigator.
Request	$18,568 for 12 months, January 1 to December 31, 2027. No award funds reach an external organization; the imaging of Aim 3 uses WSU's own C-arm. The PI leads no other AIDaS proposal this cycle [DATA NEEDED: confirm].

Abstract. Wrong-level spine surgery persists because the level is confirmed on an intraoperative C-arm image by counting vertebrae, and the count fails on transitional anatomy and short fields of view. The Open Spine Consortium at Wayne State has built the parts of an automated answer: 802 CTs with every vertebra labeled, a physics renderer that turns them into C-arm images, and a level model that names 96.4 % of 617 lateral vertebral bodies on a public C-arm dataset with no C-arm frame in its training. Two things keep it from becoming a device: its detector finds only 80.5 % of vertebrae on a C-arm it never trained on, and closing that gap takes hundreds of GPU jobs that a laboratory writes and repairs by hand. This seed project builds three linked advances. Aim 1 builds a self-repairing agent that writes, submits and repairs its own Slurm jobs on the WSU grid under gates registered before each job runs, and benchmarks it on 40 tasks against the same model in one shot. Aim 2 puts that agent in the hands of eight medical trainees through Beethoven, the laboratory's browser extension, and measures what they gain against the published 2026 cohort. Aim 3 closes the cross-device gap with randomized physics renders and a differentiable 3-D Gaussian X-ray renderer, adds corner keypoints labeled by projecting the CT, measures latency per exposure, and tests the whole stack on a paired CT and C-arm cadaver set imaged at WSU. Deliverables within the year: the agent and its benchmark under the MIT license, the cohort's outcomes, the cadaver set, an FDA Pre-Submission, and NSF CAREER and NIH R21 submissions.

Keywords. agentic systems; code generation; high-performance computing; medical education; fluoroscopy; 3-D Gaussian splatting; domain transfer; spine surgery

Project narrative

1. Significance and innovation

Surgeons confirm the level of a spine operation on an intraoperative C-arm image by counting vertebrae from a landmark, and the count fails often enough to matter. A 27-hospital Japanese registry of 34,544 degenerative spine operations recorded wrong-level surgery in 0.12 % of them, 0.42 % in the thoracic spine [1]; about half of responding US spine surgeons report at least one wrong-level operation in their career [2,3], 17 % of the events they reported led to legal action or settlement [2], and the median malpractice payment for a surgical never event is $75,220 [4]. Transitional anatomy defeats the count: 12.9 % of 550 surgical patients had a lumbosacral transitional vertebra, and their vertebrae were miscounted in clinic in 54 % of them [5]. The best-studied aid, LevelCheck, registers the preoperative CT to each image in 4.7 s on a 2012 graphics card and in 17 to 72 s in an operating room [6,7], inside or at the edge of the 20 s to one minute that surgeons said they would wait [8].

The Open Spine Consortium at Wayne State has built the parts of a faster answer: CTSpinoPelvic1K, an open dataset of 802 CTs with every vertebra, the sacrum and the hips labeled [9]; a physics renderer that turns them into C-arm images; and a detector and level model (Figure 3a) that name the level on a public C-arm dataset (Figure 4a). Three obstacles remain. The detector does not yet carry to a new C-arm: on a device held out of its training it finds 80.5 % of vertebrae, short of the 97 % target, and renders alone find almost none (Figure 4b). Closing that gap takes hundreds of GPU jobs written, refused and repaired on a shared scheduler, which is where a laboratory's time goes. And medical trainees, who hold the clinical question, cannot write those jobs.

One loop answers all three (Figure 1). **Aim 1** builds an agent that writes, submits and repairs its own jobs on the WSU grid. Its methodological contribution is a *gate-conditioned repair loop with the scheduler as part of the runtime*: a refused submission, a wrong quality-of-service string, an out-of-memory kill and a Python traceback are one class of exception, read from the scheduler's message or the job's log and repaired in the job script, and no result counts until its gate passes, a gate being a pass/fail criterion written down and hashed before the job runs. Agent benchmarks such as SWE-bench [10] score code edits on one machine; none we know of scores an agent against a shared scheduler under a compute budget and pre-registered gates. **Aim 2** gives the agent to medical trainees through Beethoven, the laboratory's browser extension that starts a trainee's Claude Code worker as a job on the grid (Figure 2). **Aim 3** spends the agent's throughput on the transfer problem, with physics renders, a differentiable 3-D Gaussian X-ray renderer [11,12] built with the PI's Gaussian-splatting methods, and a paired CT and C-arm cadaver set imaged at WSU.

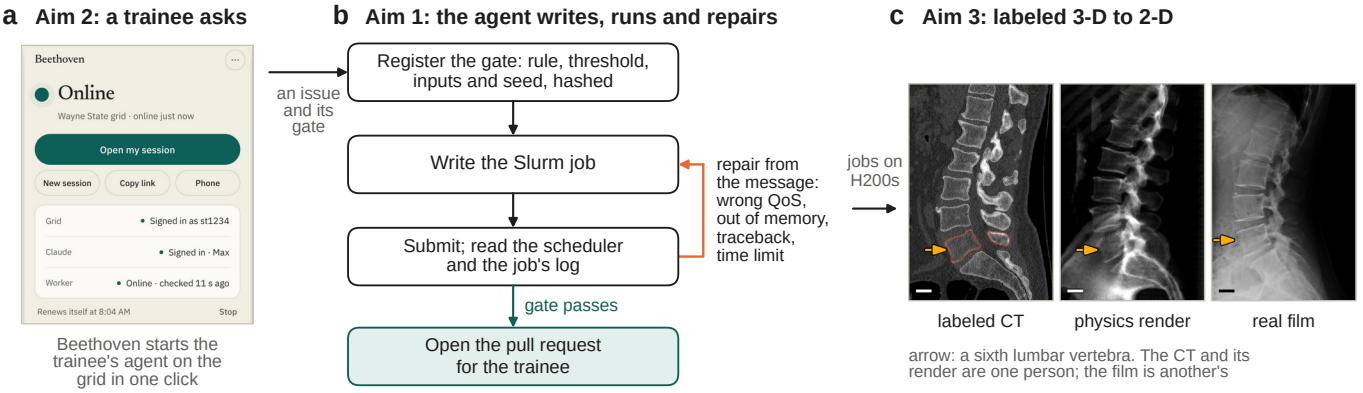


Figure 1. One loop in three aims. (a) Beethoven starts the trainee's agent on the WSU grid. (b) The agent registers the gate, writes and submits the Slurm job, and repairs from the scheduler's or the job's own message until the gate passes. (c) A sixth lumbar vertebra (arrow) in a labeled CT, its physics render, and another person's film.

2. Aims, approach and preliminary work

Aim 1 (Foundational AI): a self-repairing agent for the grid. We build and benchmark a large-language-model agent (Claude Code at a pinned model version, in a model-agnostic harness released under the MIT license) that reads the repository, writes Slurm jobs, submits them on the WSU grid and repairs refusals and exceptions with no intervention, an intervention being any human edit or resubmission. *Approach.* The agent runs the loop of Figure 1b, one issue, one branch and one pull request at a time. The benchmark holds 40 tasks from Aim 3's experiments and the repository's task lists (render banks, detector fine-tunes, figure rebuilds), each with its gate: 10 for development, 30 sealed. Three arms share one model and one budget per task (six repairs, two GPU-hours, one queued job): one shot, the repair loop, and the loop with gates; three seeds make 360 runs. *Preliminary work.* The laboratory's workers already run as Slurm jobs through Beethoven; the skeleton delivered with this proposal runs the loop against a simulated scheduler with the grid's partitions and refusals, a design check and not a result. *H1:* with the loop and its

gates the agent passes at least 80 % of the 30 sealed tasks with no intervention and no rule violation; the same model in one shot passes under half.

Aim 2 (Education and Workforce): medical trainees drive the agent. *Approach.* Eight medical students and residents in two waves of four [DATA NEEDED: confirm] each receive a grid account, Beethoven and one sub-project from Aim 1's task list with its gate; the agent writes and repairs the jobs, and the trainee owns the question, the result and the pull request, which a mentor reviews. The 2026 instrument (six comfort domains on a five-point scale) is given before and after each wave, under the 2027 cohort's own IRB determination [DATA NEEDED: HPR number]. *Preliminary work* [13]: in 2026, 16 students, seven with no coding and eleven with no imaging experience, worked on ten sub-projects with Claude as a chat coding assistant and a Slurm tutorial, but no agent that ran their jobs; on retrospective before-and-after ratings all six domains improved (Figure 2c), all ten sub-projects were submitted as abstracts to a national meeting, and all 16 would recommend the program. *H2:* all six domains improve (one-tailed Wilcoxon), and the median trainee's first successful GPU job comes within seven days.

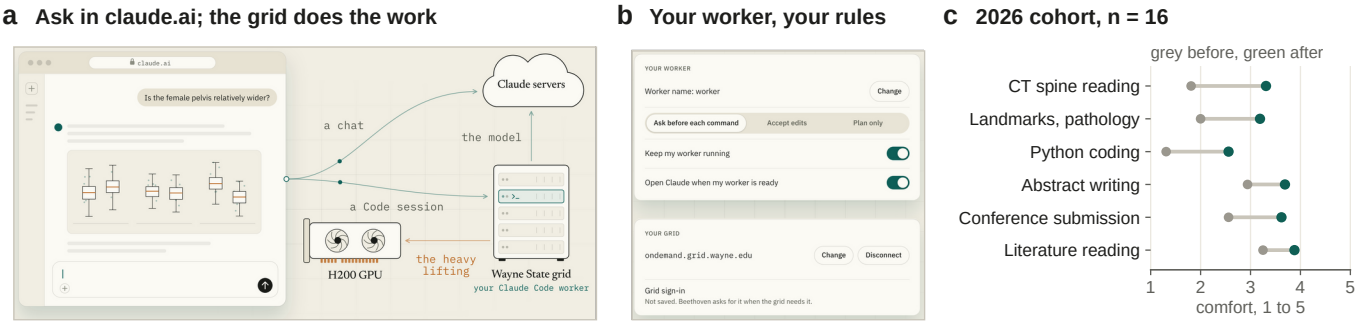


Figure 2. Beethoven, the trainee's door to the grid. (a) The trainee asks in claude.ai; the Claude Code worker runs as a Slurm job on the grid, whose H200 partition has 4 nodes of 4 GPUs. (b) The trainee sets the worker's rules. (c) The 2026 cohort's comfort before and after (n = 16; every domain improved, one-tailed Wilcoxon $p < 0.05$) [13].

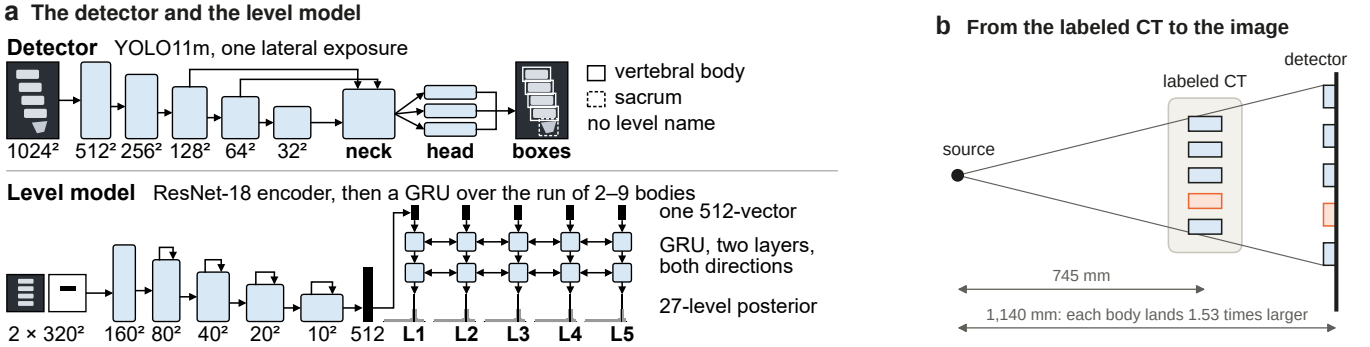


Figure 3. From the labeled CT to the C-arm image. (a) A YOLO11m detector boxes the bodies; a ResNet-18 encoder and a two-layer bidirectional GRU name each one among 27 levels. (b) At the device's geometry (0.1915 mm per pixel) the projection carries each labeled body of the CT, here a transitional one in orange, to its place on the image.

Aim 3 (AI for Health): labeled 3-D structure into 2-D fluoroscopy, at low latency. *Data.* Physics renders of CTSpinoPelvic1K, VerSe and TotalSegmentator CTs (7,045 standard and 1,482 transitional annotated frames) and a device-styled corpus of 142 TotalSegmentator spines (12,730 entries); real frames of three devices: C-ArmLSpine (149 cases; 617 lateral and 631 AP bodies whose truth is automatic, two CBCT labelers agreeing), DeepFluoro and COBRA Elekta. *Model* (Figure 3a): a YOLO11m detector boxes each body and the sacrum; a ResNet-18 encoder and a bidirectional GRU name each body; a count from the sacrum checks the names, and the stack abstains where count and shape disagree. New here: four corner keypoints per body, labeled for free by projecting each CT body's corners at the device's geometry (Figure 3b). *Closing the gap.* Physics renders randomized over devices' geometry and spectra; real frames of two devices with the third held out; and a differentiable 3-D Gaussian X-ray renderer fitted to each labeled CT, whose spectrum, scatter and detector response are learned until its renders stop being separable from real frames, which today styled renders are 99 to 100 % of the time. *Cadaver set.* Two specimens from WSU's Body Bequest Program are CT-scanned, then imaged AP and lateral at the protocol's lumbar and thoracic positions on WSU's C-arm in Room 0312 Elliman (booked through DLAR, with OEHS dosimetry); a frame grabber records the C-arm's video output, the set is sealed by hash before any model sees it, and the CT gives every frame its truth. *Preliminary results* (Figure 4, all held out by person): with no C-arm frame in training the level model names 96.4 % of 617 lateral bodies on C-ArmLSpine but 43.7 % of AP bodies, and the device's own frames, from other patients, lift AP to 96.4 %; count and shape together name 98.2 % of the 1,009 bodies on frames that show the sacrum, and a stop

signal fires on 9.9 % of 233 anchored frames, catching five of the six that err. Trained on two other devices' real frames, the detector finds 80.5 % of C-ArmLSpine's vertebrae at the 0.74 cross-device operating point; no held-out cell met the pre-stated 97 %. The stack's end-to-end latency has not been measured; Aim 3 measures it first. *H3:* 97 % of vertebrae detected on the held-out device at 0.74, under 1 s per exposure on one H200, and every cadaver frame reported against its CT-derived truth.

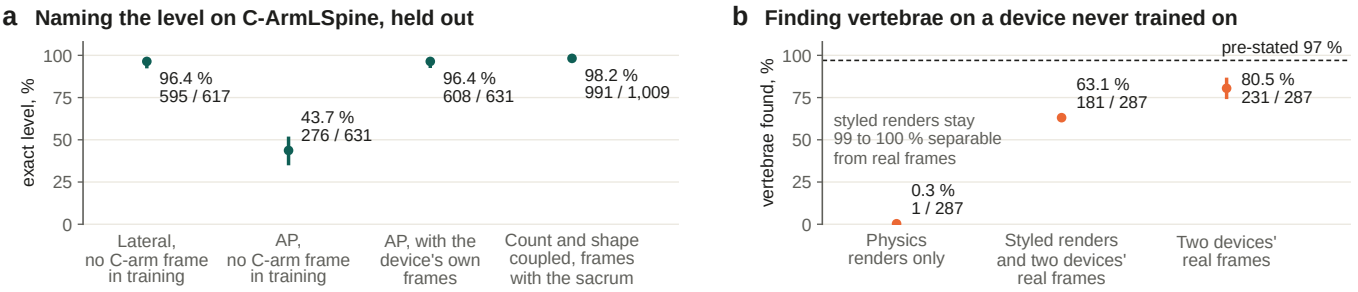


Figure 4. Held-out results with 95 % Wilson intervals on the effective sample size where the record gives one. (a) Exact level named on C-ArmLSpine, without and with the device's own frames in training. (b) Vertebrae found on C-ArmLSpine, held out of detector training, at the 0.74 cross-device operating point, against the pre-stated 97 %; at 0.25 the two-device detector finds 90.6 %.

3. Evaluation: data, metrics and design

Aim	Data	Metrics	Design	Gate, written before the run
1	40 tasks from the repository and Aim 3, each with its gate: 10 development, 30 sealed	pass rate; repairs per task; interventions; rule violations (a login-node run, a wrong QoS, a credential in a file); GPU-hours per pass	three arms, one model, one budget per task; three seeds; paired bootstrap over tasks	$\geq$80 % pass on the 30 sealed tasks, 0 interventions, 0 violations; one shot passes under half
2	2027 cohort, 8 trainees in two waves [DATA NEEDED: confirm]; the 2026 cohort (n = 16) as context	six-domain comfort before and after; days to first successful GPU job; tasks passed without a mentor; pull requests and abstracts	prospective cohort, the 2026 instrument, analysis plan registered before enrollment	all six domains improve (one-tailed Wilcoxon, $p < 0.05$); median first GPU job within 7 days
3	C-ArmLSpine, DeepFluoro and COBRA Elekta frames; the render banks; the sealed cadaver set	exact level per frame; vertebra and sacrum recall at 0.74; corner error (mm); Dice and IoU of the projected silhouette; latency per exposure (median, p95); abstention; render-to-real separability	leave-one-device-out; the cadaver set opened once, after code and weights are hashed	97 % detection on the held-out device; under 1 s per exposure on one H200; every cadaver frame reported against CT truth

Frames cluster within cases, so every interval is Wilson on the effective sample size. The cadaver set is a sanity test on real anatomy with CT truth, not a powered test; the powered tests are the held-out device and C-ArmLSpine, and the laboratory's power analysis (35 persons with no error for a 90 % lower bound) sets the R21's sample.

4. Expected outcomes, deliverables and timeline

Within the award: (1) the agent, its benchmark and the three-arm report, under the MIT license with the tasks a disclosure review clears; (2) the cohort's outcomes, pull requests and abstracts; (3) the paired CT and C-arm cadaver set, deposited beside CTSpinoPelvic1K as the Body Bequest Program's terms allow, with the transfer and latency results in a manuscript; (4) an FDA Pre-Submission; (5) NSF CAREER and NIH R21 submissions. *Quarter 1:* agent version 1, benchmark sealed, IRB, recruitment, specimens and C-arm booked. *Quarter 2:* benchmark run, wave 1 (months 4 to 7), cadaver imaging, latency and transfer results; Pre-Submission filed in April, so that FDA's written feedback, due within 70 days, precedes the R21. *Quarter 3:* CAREER (July 28), wave 2 (months 8 to 11), Gaussian renderer. *Quarter 4:* R21 (October 18), cadaver set opened, final report and AIDaS symposium. *Risks.* An agent not ready by month 4 starts wave 1 on the 2026 setup; a delayed specimen leaves the held-out device's frames as the test; a transfer result under 97 % becomes the R21's first aim; each task's budget caps its cost.

References. [1] Furuya M, et al. Bone Joint J 2026;108-B(5):673-80. [2] Mody MG, et al. Spine 2008;33(2):194-8. [3] Groff MW, et al. World Neurosurg 2013;79(3-4):585-92. [4] Bettis T, et al. Patient Saf Surg 2026. doi:10.1186/s13037-026-00508-9. [5] Nakagawa T, et al. Tohoku J Exp Med 2017;242(3):223-8. [6] Otake Y, et al. Phys Med Biol 2012;57(17):5485-508. [7] Manbachi A, et al. Ann Biomed Eng 2018;46(10):1548-57. [8] De Silva T, et al. Spine 2016;41(20):E1249-56. [9] Open Spine Consortium. CTSpinoPelvic1K, release 11. Zenodo, doi:10.5281/zenodo.22797014. [10] Jimenez CE, et al. SWE-bench: can language models resolve real-world GitHub issues? ICLR 2024. [11] Cai Y, et al. Radiative Gaussian splatting for efficient X-ray novel view synthesis. ECCV 2024. [12] Zha R, et al. R^2-Gaussian: rectifying radiative Gaussian splatting for tomographic reconstruction. NeurIPS 2024. [13] OpenSpineConsortium: an open-source framework for medical student engagement in computational spine imaging research. Cureus 2026;18(7):e112661.

AIDaS funding priority alignment

Target agencies, programs and solicitations

1. **NSF CAREER** (NSF 22-586; CISE, Division of Information and Intelligent Systems), PI Han. Research plan: Aim 1's gate-conditioned agent and Aim 3's 3-D-to-2-D learning; education plan: Aim 2's trainees. Deadline July 28, 2027, the fourth Wednesday of July; eligibility is judged at the deadline (untenured Assistant Professor, no earlier CAREER award) [DATA NEEDED: the PI's tenure timing and earlier CAREER competitions]. If the PI is not eligible, the same plan goes to IIS's core programs (Future CoRe, NSF 25-543, Robust Intelligence).
2. **NIH R21 at NIBIB**: the reissued Trailblazer Award for New and Early Stage Investigators (forecast FOR-EB-26-001; the last notice, PA-25-169, expired May 8, 2026), or else NIBIB's Exploratory/Developmental Bioengineering Research Grant (EBRG, PAR-25-321) or the parent R21 (PA-25-304), both open through October 2027. Submission October 18, 2027, the standard October 16 falling on a Saturday, while the PI is still an Early Stage Investigator. Preliminary data: the cadaver set, the held-out-device result and the measured latency; FDA's Pre-Submission feedback shapes its validation aim.
3. **NSF CyberTraining** (NSF 23-520), a Pilot of up to $300,000 over two years led by Co-PI Schwiebert: agent-assisted training of researchers on advanced computing, from Aim 2. Deadline January 20, 2028, the third Thursday of January; submitted in any case.
4. **NSF Smart Health** (SCH; the last solicitation, NSF 25-542, awaits its successor): the whole loop as a four-year project once NSF publishes the next call.

Relevance to national science and technology priorities

America's AI Action Plan (July 2025) asks for investment in AI-enabled science and, in its worker-first agenda, for AI skills in the workforce, which Aim 2 builds in Michigan's medical trainees. Executive Order 14363 (November 2025) launches the Genesis Mission, which joins AI agents to national computing, and NSF's Dear Colleague Letter 26-023 invites proposals to existing programs that support it, as NIH's Bio Genesis Mission does for biomedicine. Aim 1 is that agenda at a university's scale: an agent that runs science on shared computing and is held to verifiable gates. Executive Order 14303, Restoring Gold Standard Science (May 2025), and OSTP's guidance of June 2025 ask agencies to promote pre-registration of study protocols and transparent reporting of null results; every gate in this project is pre-registered by hash, and failed gates are reported. NIH's data sharing policy and its Strategic Plan for Data Science 2025–2030 ask for data others can reuse, which CTSpinoPelvic1K and the cadaver set are, and FDA's Q-Submission program gives Aim 3 a regulatory path within the award.

Team, expertise and roles

The PI, a computer vision researcher who leads the Machine Perception Lab and publishes on 3-D Gaussian splatting and neural implicit surfaces (CVPR 2025 and 2026, NeurIPS 2024), builds the agent and the 3-D-to-2-D model, including the Gaussian X-ray renderer. The Co-PI, a high-performance computing researcher who accelerates scientific codes on GPUs, builds the benchmark harness and the grid integration and leads the CyberTraining Pilot; the Co-PI and the clinical collaborator have published four high-performance computing papers together (2021 to 2024). The clinical collaborator holds the record this proposal builds on and runs the cohort, the IRB determination and the imaging session. The part-time researcher of line A runs the benchmark, logs the measures and curates the cadaver set. Effort: PI [DATA NEEDED: %], Co-PI [DATA NEEDED: %], in kind.

Timeline for external submission

When	Milestone
January to March 2027 (months 1–3)	One-page concepts to the NSF IIS and NIBIB program officers; benchmark and gates sealed
April 2027 (month 4)	FDA Pre-Submission filed; written feedback due within 70 days
July 28, 2027 (month 7)	NSF CAREER submitted, with Aim 1's benchmark and Aim 3's leave-one-device-out results
October 18, 2027 (month 10)	NIH R21 submitted, with the cadaver set, the latency and FDA's feedback
December 2027 (month 12)	Project ends; AIDaS final report and symposium
January 20, 2028 (post-award month 1)	NSF CyberTraining Pilot submitted, Co-PI leading
2028	R21 resubmission at the first standard date after its summary statement, under the parent R21 that replaces PA-25-304 [DATA NEEDED: the reissued notice]; CAREER resubmission July 26, 2028, if eligible; SCH when NSF publishes it

The CAREER and R21 meet the expectation of a federal proposal within 12 months of the project's end, and the CyberTraining Pilot follows whatever their outcome; publications and proposals from the award acknowledge AIDaS.

Budget and justification

Line	Item	Basis	Amount
A	Part-time researcher, hourly	520 h at $15.00: Aim 1 180 h, Aim 2 160 h, Aim 3 180 h	$7,800
A2	Fringe on line A	FICA only, 7.65 %	$597
B1	Two cadaver specimens, WSU Body Bequest Program	placeholder $1,250 each [DATA NEEDED: program fee]	$2,500
B2	Specimen CT, two scans	placeholder $300 each [DATA NEEDED: facility and rate]	$600
B3	C-arm session, Room 0312 Elliman, through DLAR	4 h at a placeholder $300 [DATA NEEDED: DLAR rate]	$1,200
B4	Dosimetry and OEHS review for the session	placeholder [DATA NEEDED: OEHS fee]	$300
B5	Frame grabber for the C-arm's video output	Magewell USB Capture AIO, list price	$619
C1	Claude Team Premium seats for the agents	6 seats at $15 a month for 12 months	$1,080
C2	Data management and storage	2 TB of C&IT hosted storage at $400 per TB-year; Zenodo at no charge	$800
D	FDA Pre-Submission: regulatory consultant	10 h at $300; FDA charges no fee	$3,000
E	Anthropic AI for Science application	no fee; 8 of Aim 1's hours in line A	$0
F	AIDaS symposium poster	36 by 48 in at $6 per sq ft	$72
	Total requested		**$18,568**

- **A, A2.** An hourly graduate student assistant where one is available, which is the call's graduate student support, or else a post-baccalaureate researcher; ten hours a week for 52 weeks. $15.00 is Michigan's minimum wage from January 1, 2027 (MCL 408.934) [DATA NEEDED: WSU's job category and scale]. Aim 1's hours are 360 runs at half an hour each; Aim 2's run the cohort's accounts and measures; Aim 3's are 100 hours curating the cadaver set (de-identifying, labeling, sealing by hash) and 80 running the leave-one-device-out and latency jobs. Temporary employees are not benefit-eligible, so the fringe is FICA alone (an enrolled student owes less). At $18 an hour the total would be $20,247, over the cap unless the hours fall to 507.
- **B1 to B5.** A paired CT and C-arm set of the same anatomy exists in no public dataset, and it is the only test in which an X-ray's truth comes from a CT rather than a reader. The donor authorization of the Body Bequest Program permits de-identified images [DATA NEEDED: the program's terms for public release]. The C-arm is WSU's own, in Room 0312 Elliman, booked through DLAR: every user is enrolled in WSU's dosimetry program, with badges in the hands of DLAR's director at least 15 business days before the session, and completes OEHS's training form. The frame grabber records what the device shows, which is the deployment input; it takes SDI and composite video, since the C-arm's live output may be BNC [DATA NEEDED: the C-arm's model and output]; at $619 it is far under WSU's $5,000 capital threshold and becomes WSU property.
- **C1, C2.** Anthropic's Team plan for scientists lets a verified PI add Premium seats at $15 a month: four for the trainees, shared by the two waves, and two for the benchmark and development. The plan takes card payment only [DATA NEEDED: P-card route and the PI's verification]; at Claude Pro's $20 a month the line would be $1,440. The dollars are list prices. The WSU grid's compute is free to WSU researchers and given in kind. Storage holds the render banks, the cadaver set and every benchmark run's logs for the award year; the data are deposited on Zenodo at its end.
- **D.** FDA charges no fee for a Pre-Submission filed in 2027 (a fee is proposed only from fiscal year 2028); the consultant, at a mid-level regulatory rate, reviews a package the PI writes (device description, intended use, proposed testing), so that FDA's written feedback reaches the R21.
- **E, F.** AI for Science awards up to $20,000 in API credits for six months and charges nothing to apply; credits would add benchmark seeds. Presenting at an AIDaS symposium is an award expectation; the poster is printed at the School of Medicine's Medical Communications.
- **Not requested.** PI and Co-PI salary, travel, tuition, equipment and indirect costs [DATA NEEDED: confirm AIDaS applies no F&A].
- **Allowability and dependencies.** Lines A (as graduate student support), C1 and C2 fall under the call's allowable headings and alone total $10,277, above the $10,000 floor. Lines B, D and F are research costs the AIDaS office is asked to confirm before submission. B2 to B5 depend on B1, since without a specimen there is nothing to scan or image. Without B the request is $13,349 and Aim 3 tests on the held-out device's frames, its 100 cadaver hours moving to extra benchmark seeds; without B and D it is $10,349; without B, D and F, $10,277.

Biosketches

The two investigators' biosketches, two pages each in NSF format, follow this page in the submitted PDF. Put `biosketch_han.pdf` and `biosketch_schwiebert.pdf` beside `main.tex` and compile again, and they replace this page.